FINAL GRADUATION PROJECT REPORT

(Center Align)
Alamein International University
Computer Science & Engineering

EduVR: AI-Powered Collaborative Virtual Reality Learning Platform
A project submitted
in partial fulfillment of the requirements for the BSc. degree
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UNDERTAKING

This is to declare that the project entitled **"EduVR: AI-Powered Collaborative Virtual Reality Learning Platform"** is an original work done by the undersigned, in partial fulfillment of the requirements for the Bachelor of Science (BSc.) degree at the Faculty of Computer Science & Engineering (CSE), Alamein International University (AIU).
All the analysis, design, implementation, and system development activities have been accomplished by the undersigned. Moreover, this project has not been submitted to any other college or university for the award of any degree or diploma.
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ABSTRACT

Traditional educational methods often struggle to effectively teach complex three-dimensional concepts due to the limitations of 2D screens and textbooks. These restrictions negatively impact student engagement and understanding, particularly in fields like medicine and engineering where spatial awareness is critical.

To address this, we present **EduVR**, a collaborative Virtual Reality platform designed to create an immersive and interactive learning environment. The system allows instructors and students to join shared virtual classrooms where they can explore, dissect, and discuss high-fidelity 3D models in real time, regardless of their physical location.

The backbone of this ecosystem is the **EduVR Core**, a robust cloud-based system responsible for content management and distribution. The Core currently features an **Automated Asset Processor** that optimizes raw 3D files for performance, and a **Room Manager** that orchestrates multiplayer sessions between the **Unity VR Client** and the **Web Visualization Interface**.

By establishing this scalable technical foundation, EduVR supports advanced interactions such as cross-sectional slicing and model isolation. Furthermore, the Core is architected to support the future integration of a **Retrieval-Augmented Generation (RAG) AI Tutor**, which will eventually allow students to ask questions via voice or text and receive context-aware answers derived directly from the educational material.

In summary, EduVR provides a cost-effective and scalable solution that modernizes laboratory education, enhances collaboration, and significantly improves conceptual retention across scientific domains.

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Section 1: Requirements Listing and Analysis

Requirements Analysis

This section defines the system requirements of the EduVR platform based on the identified educational needs and problem analysis. The requirements are classified into functional and non-functional requirements to clearly describe the expected behavior and quality attributes of the system.

Functional Requirements

The following table outlines the core functions available to Instructors and Students within the EduVR ecosystem.

Table 1: Functional Requirements of the EduVR System

ID	Requirement Description
FR1	The system shall allow instructors and students to join shared virtual learning sessions.
FR2	The system shall support real-time multi-user interaction within a virtual environment.
FR3	The system shall allow instructors to control and manipulate shared 3D models.
FR4	The system shall allow students to view and interact with 3D models.
FR5	The system shall provide an AI Smart Tutor (EduVR Core) to answer user questions.
FR6	The system shall support text and voice-based interaction with the AI tutor.

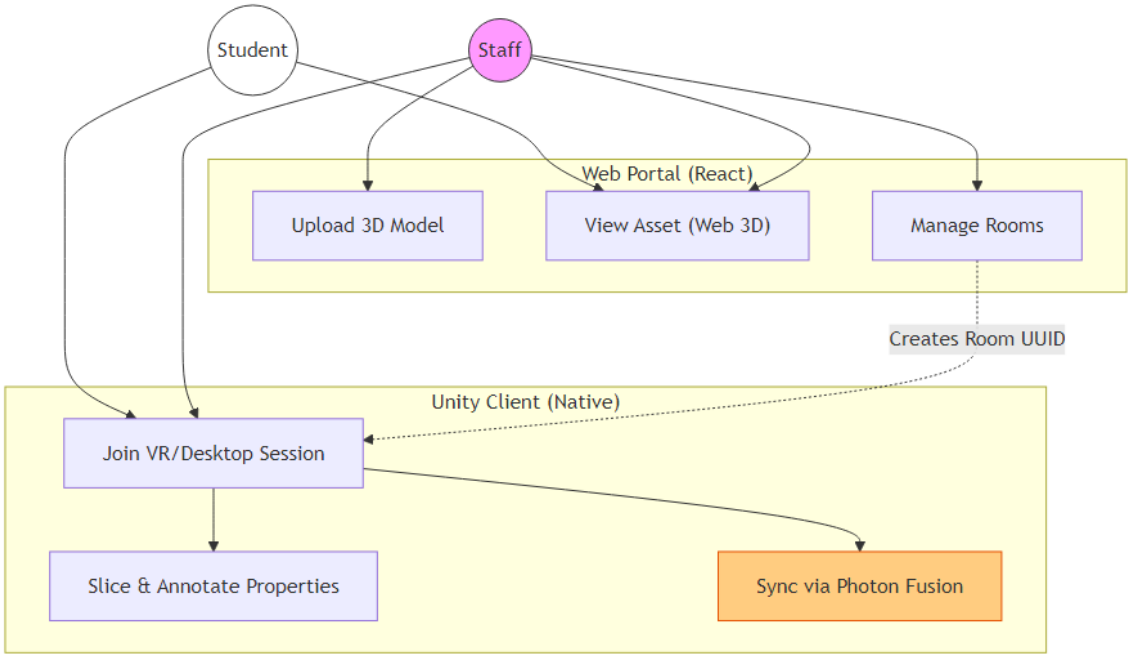

Figure 1: Use Case Diagram for EduVR

Non-Functional Requirements

The following table outlines the key quality attributes and performance constraints for the VR Learning Platform.

Table 2: Quality Attributes and Performance Constraints

ID	Category	Requirement Description
NFR1	**Performance**	The VR application must maintain a minimum of 72 FPS on target hardware (e.g., Meta Quest 2/3) to prevent motion sickness.
NFR2	**Latency**	Network latency for multi-user synchronization (movement, voice, object interaction) should not exceed 100ms.
NFR3	**Scalability**	The backend must support concurrent processing of uploaded 3D models without degrading the performance of active VRsessions.
NFR4	**Reliability**	The **EduVR Core** service should have 99% uptime during active learning sessions.
NFR5	**Usability**	The VR interface must be intuitive enough for a user to join a room and interact with an object within 2 minutes of first use.

Section 2: Solution Design and Diagrams

Solution Design

The EduVR solution is designed as a collaborative learning platform that integrates Virtual Reality, Artificial Intelligence, and real-time networking technologies. The system design focuses on modularity and scalability to allow future extensions and support different educational domains.
The system architecture is divided into user interfaces (VR & Web), backend services, the **EduVR Core** (AI Service), and data storage components.

System Architecture

The high-level architecture demonstrates the interaction between the Unity Client, the Central Backend, and the **EduVR Core**.

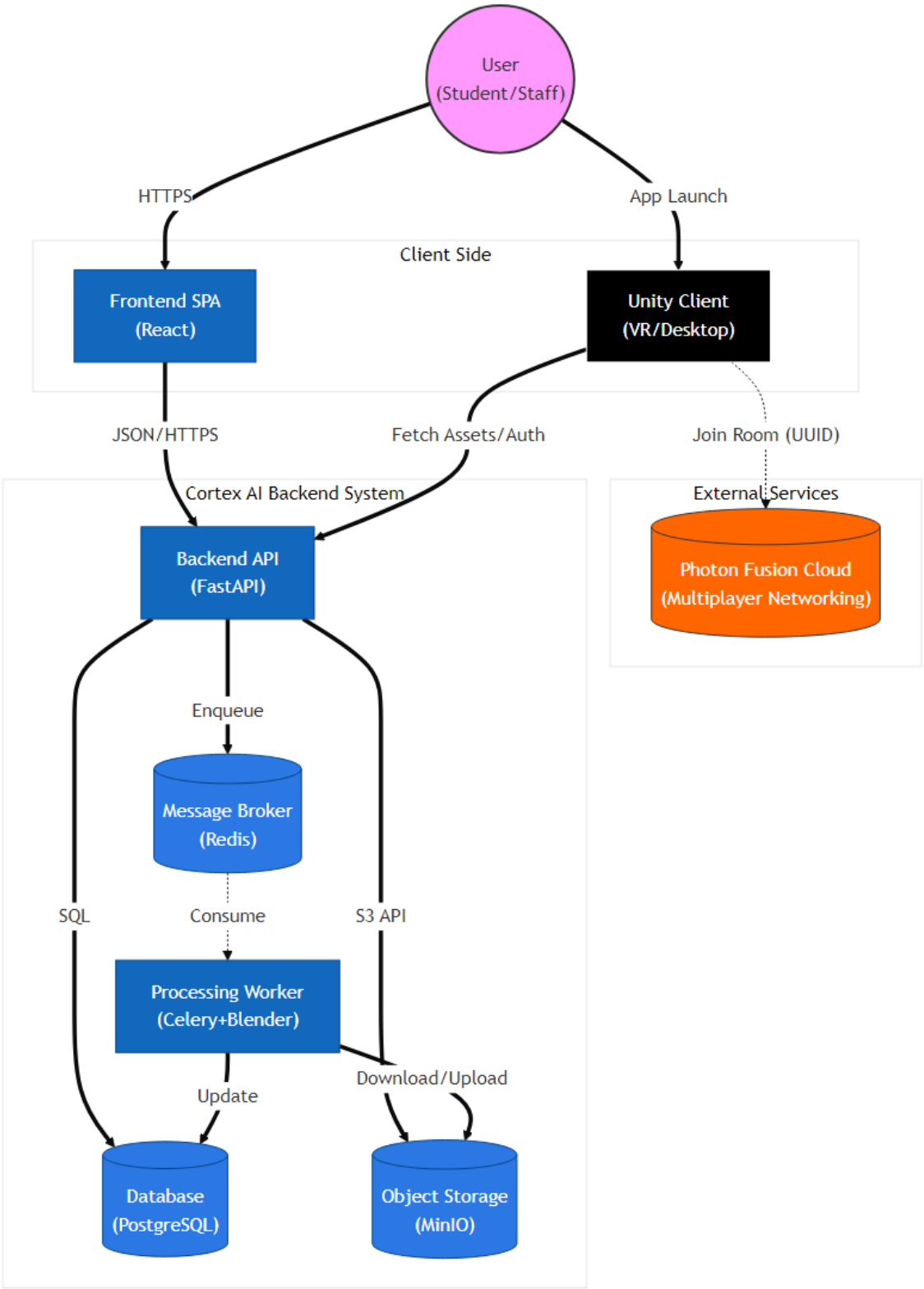

Figure 2: System Architecture Diagram

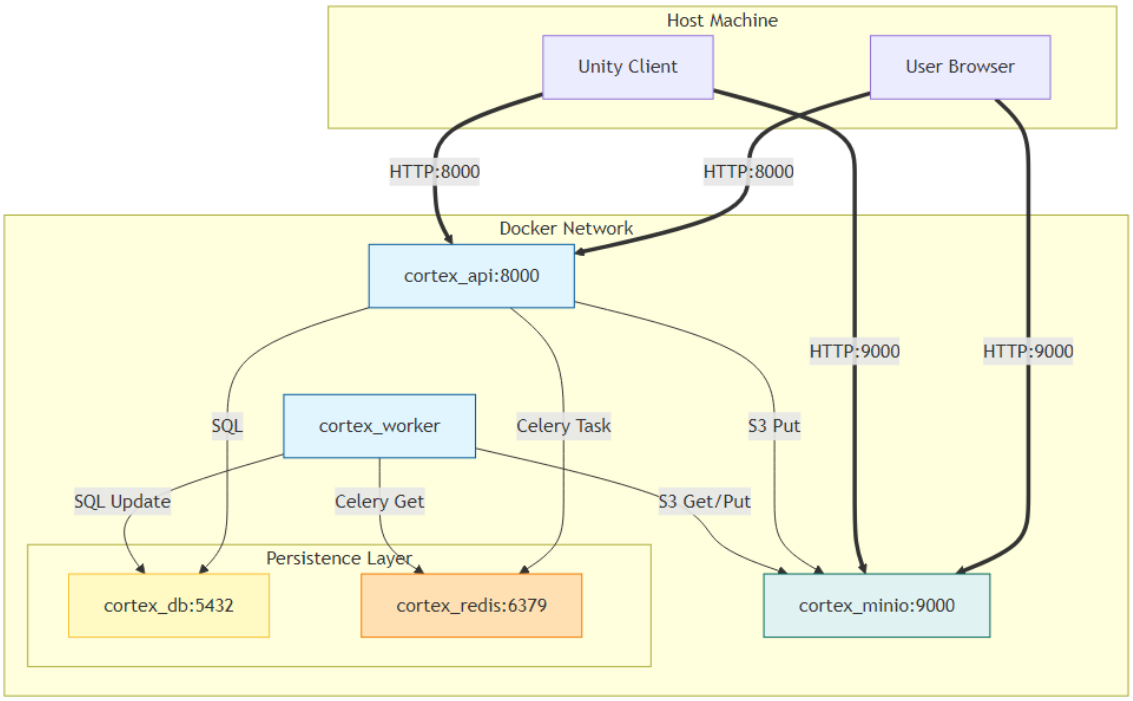

Figure 3: Infrastructure and Deployment Diagram

System Workflows

The following sequence diagrams illustrate how users interact with the system, specifically the process of joining a room and how 3D assets are processed by the backend.

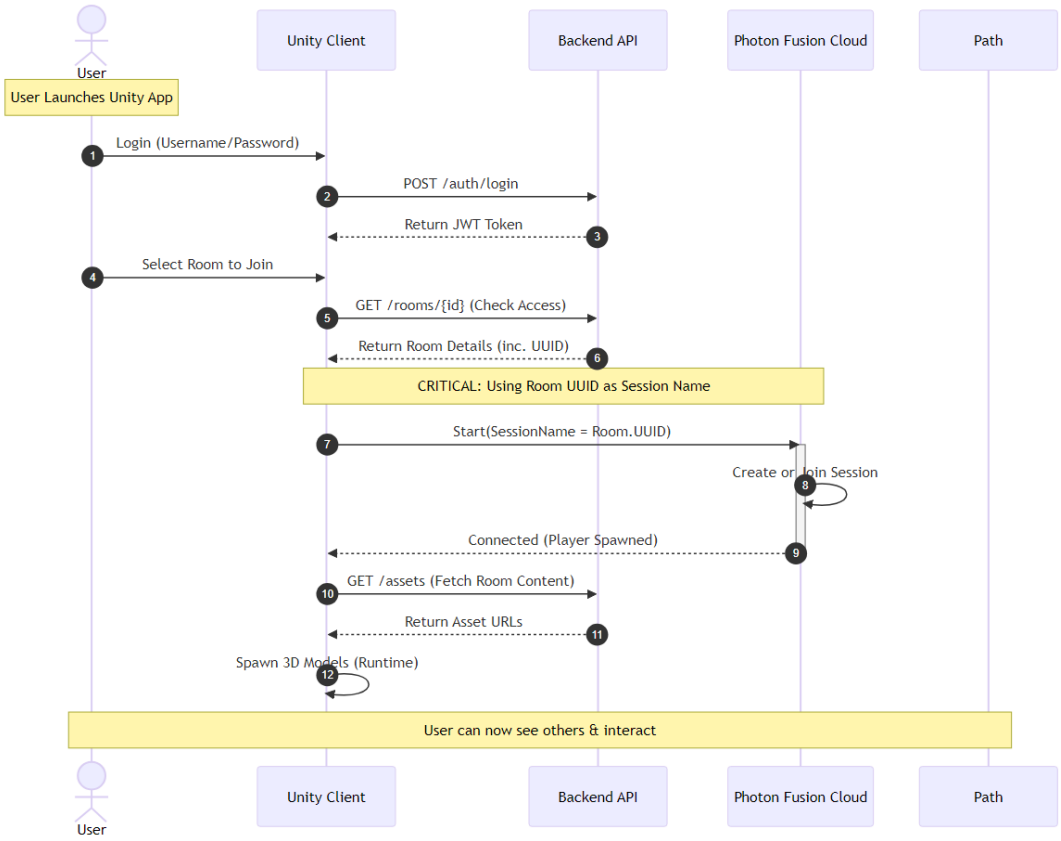

Figure 4: Room Join Sequence Diagram

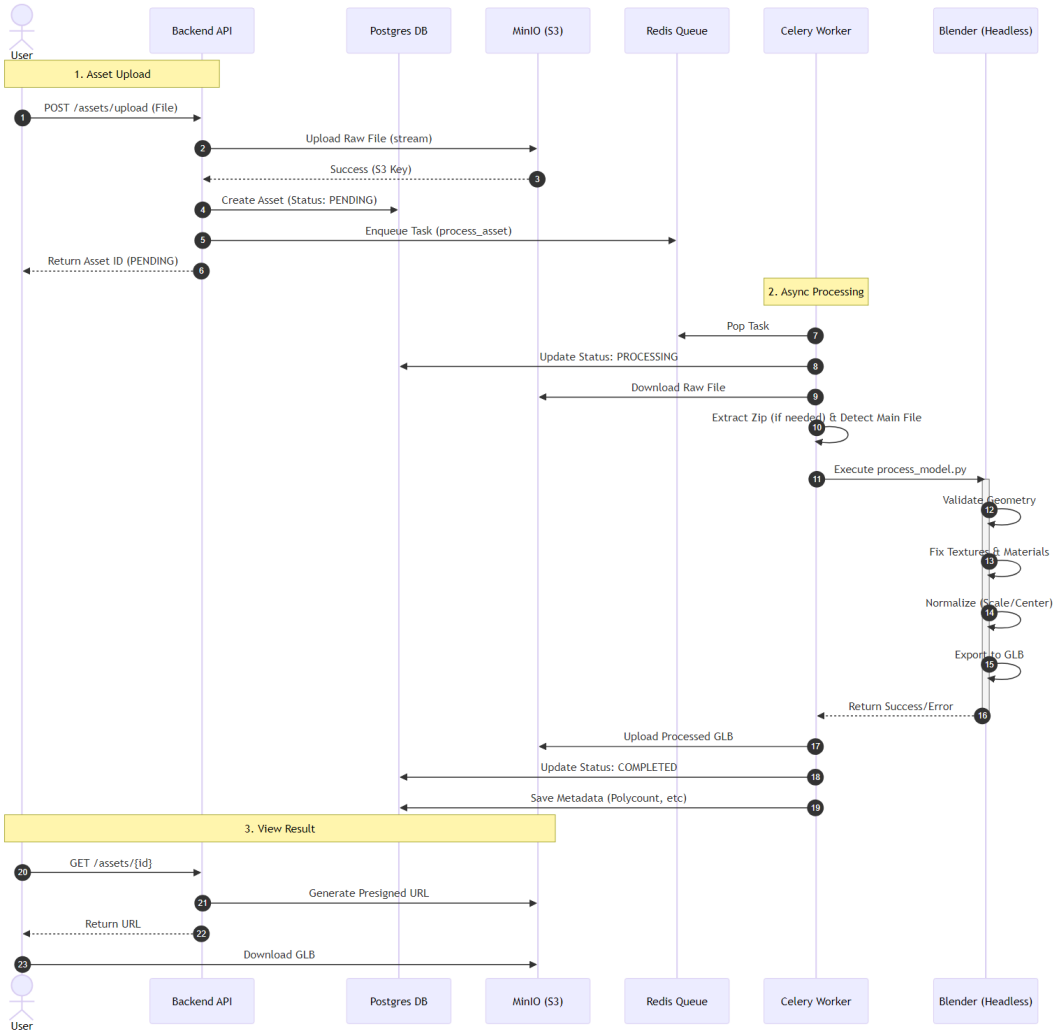

Figure 5: 3D Asset Processing Sequence Diagram

Class Structure

The internal code structure for the EduVR application is organized as follows:

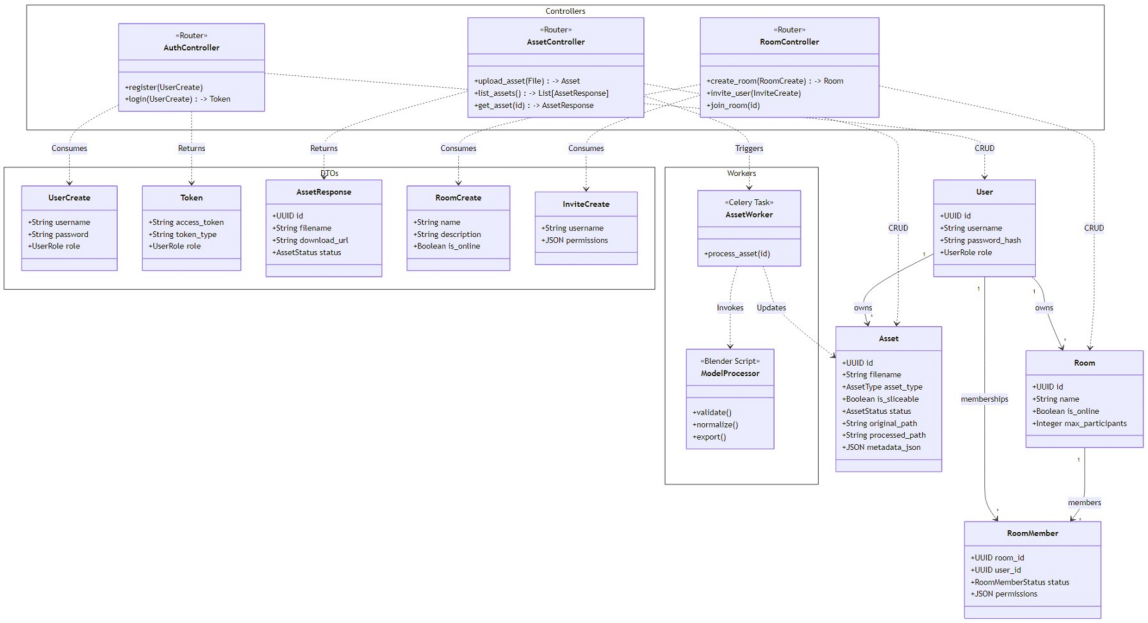

Figure 6: Class Structure Diagram

Section 3: Project Planning

Project Planning

The development of the EduVR project followed a structured software engineering approach to ensure timely delivery and system quality. The project was divided into several phases, including requirements analysis, system design, implementation, testing, and evaluation.

The initial phase focused on defining project objectives and selecting appropriate technologies. The design phase involved planning the system architecture and interaction flow. The implementation phase focused on developing the VR environment, backend services, and the **EduVR Core** components. Finally, testing and evaluation were conducted to validate system functionality and performance.

Team roles were distributed among members to ensure efficient collaboration. Each team member was responsible for specific tasks such as development, documentation, research, and testing.

The following Gantt chart illustrates the timeline of the project from inception to final delivery.

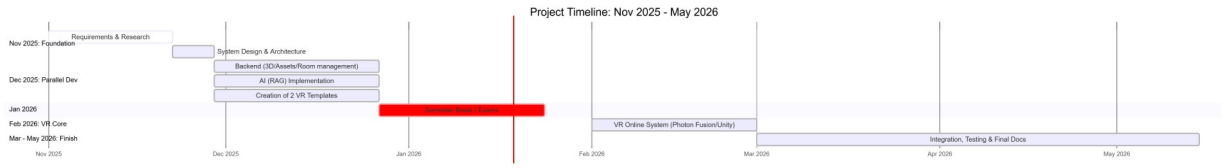

Figure 7: Project Development Timeline

Future Work (Phase 2)

As outlined in the timeline, the upcoming academic term will focus on completing the full system ecosystem. Key objectives for this next phase include:

1. **Online Unity System:** Finalizing the robust multi-user networking layer within Unity.
2. **AI Builder Assistant:** Developing a tool to assist instructors in creating 3D content effortlessly.
3. **RAG-Powered AI Tutor:** Fully connecting the AI Tutor interface to the main backend **EduVR Core**, enabling context-aware responses based on educational material.

Section 4: Implementation Approach

Implementation Approach

The implementation of EduVR focused on integrating Virtual Reality interaction, multi-user collaboration, and AI-based assistance into a unified system. Unity was used to develop the VR environment and support user movement and object interaction. Real-time collaboration was implemented to synchronize user actions and shared objects.
Backend services were implemented to handle authentication, session management, and communication between system components. The **EduVR Core** was implemented using a Retrieval-Augmented Generation (RAG) approach, where user queries are processed by retrieving relevant information from verified sources before generating responses.
An automated pipeline was implemented to process and optimize 3D assets for use across both VR and web platforms. This approach ensured consistent performance and reduced manual preparation effort.

Current Phase Deliverables

To demonstrate the full scope of the system's capabilities, we have delivered both ready-to-use educational content templates and the core technical infrastructure that powers them.

1. Educational Content Templates

Two immersive environments have been prepared for immediate use:

- **Anatomy Template:** A highly detailed immersive environment for medical education, utilizing the open-source Z-Anatomy atlas [7] to provide medically accurate 3D models for dissection and study.
- **Mathematics Visualization Template:** An interactive environment designed for visualizing complex mathematical functions and geometric concepts in 3D space.

2. Core Technical Modules

In addition to the content, three foundational software modules have been fully implemented and integrated during this phase:

- **Automated 3D Model Processor:** A robust backend pipeline (utilizing Celery and Headless

Blender) that automatically ingests raw uploaded files (GLB, FBX, ZIP). It performs critical "sanity checks"—validating geometry, repairing broken texture links, fixing transparency artifacts, and optimizing the mesh—ensuring that every model runs smoothly on Web and VR clients without manual intervention.

- **Asset Management System:** A centralized web interface that allows Staff users to upload, organize, and manage their 3D library. It enforces strict Role-Based Access Control (RBAC) to ensure content security, only permitting authorized staff to introduce new assets into the ecosystem.
- **Collaboration Room Manager:** A dedicated subsystem for orchestrating multi-user sessions. It enables instructors to create virtual classrooms, control their availability (Online/Offline), and manage student access via an invitation system. This module serves as the critical bridge between the web platform and the multiplayer environment, generating the unique Session IDs used by the Photon Fusion networking engine.

Section 5: Adopted Ethics Principles

Adopted Ethics Principles

The EduVR project was developed in accordance with recognized software engineering ethics principles. The primary ethical principles adopted include user privacy, data security, transparency, and responsible use of Artificial Intelligence.

User Privacy & Data Security:

User data and interactions are handled securely, and role-based access control is implemented to prevent unauthorized actions. User privacy is respected by limiting data collection to essential information required for system operation.

Responsible AI (EduVR Core):

The EduVR Core follows responsible AI practices by using verified knowledge sources to reduce incorrect or misleading information. Ethical alternatives such as using unrestricted AI models were considered; however, a Retrieval-Augmented Generation approach was chosen to ensure accuracy and reliability in an educational context.

Licensing & Copyright:

All software tools, libraries, and frameworks used in the project follow their respective licenses and usage policies. Open-source tools were used in compliance with their licenses, and no copyrighted content was used without permission.

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